

Michael Melichar

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EDUCATION

University at Buffalo

Bachelor of Science in Computer Science

Buffalo, NY

Aug. 2024 – Expected Dec. 2027

EXPERIENCE

UB Nanosatellite Laboratory | Flight Software Engineer

Sept. 2025 – Present

University at Buffalo

Buffalo, NY

- Write flight software in C on NASA core Flight System (cFS), handling telemetry and command messages routed over the cFS Software Bus.
- Built test applications in C that send health packets over UART through the EyeStar radio to verify connectivity with the Globalstar satellite network.
- Integrated message passing between the flight computer and an onboard NVIDIA Jetson GPU using the lab's payload communication API.
- Bench-tested sensor and radio hardware on Linux-based flight computers, debugging serial links and confirming packet integrity end to end.

UB SEDS | Avionics Software Engineer

Aug. 2024 – Present

University at Buffalo

Buffalo, NY

- Wrote embedded firmware in C/C++ on an STM32 running FreeRTOS to drive solenoid valves for loading nitrous oxidizer into a hybrid rocket during ground fill operations.
- Programmed a stepper-motor driver to retract the rocket's umbilical line at launch, and built the igniter firing driver.
- Built abort and safety task handling as FreeRTOS tasks that monitor fault conditions and trigger a controlled shutdown of the fill and ignition sequence.
- Prototyped a Kalman filter in C++ for IMU-based state estimation, validated against simulated accelerometer and gyroscope data.

Linde | Software Engineering Intern

May 2026 – Aug. 2026

Buffalo, NY

- Built a customer-facing web app in C# and Blazor that visualizes live and historical readings from cryogenic freezers, so customers can track temperature and fill levels directly.
- Wrote data pipelines that pull historical weather data from public web APIs into Microsoft Fabric, feeding a forecasting model covering 8,000+ plant locations.
- Modeled and structured the ingested data in Microsoft Fabric to support reporting and downstream analysis.

PROJECTS

Create: Civilization (Founder & Owner) | Java, Kotlin, Python, React, SQLite

June 2024 – Present

- Founded and run a modded Minecraft server network that has grown to 680+ users at 99.8% uptime, including load balancing and performance monitoring.
- Wrote server-side mods in Java with custom event systems and performance fixes that hold up under sustained player load.
- Built and maintain the community website in React and Discord moderation and alert bots in Python over REST APIs.
- Run the team on Scrum and wrote Python scripts for player onboarding, automated backups, and log parsing.

TECHNICAL SKILLS

Embedded / Systems: NASA cFS, FreeRTOS, STM32, Linux, UART/Serial, GPIO, Raspberry Pi, NVIDIA Jetson

Languages: C/C++, C#, Python, Java, Kotlin, JavaScript, SQL (Postgres, MySQL, SQLite), HTML/CSS

Web / Frameworks: Blazor, .NET, React, Node.js, Flask, FastAPI

Tools / Cloud: Git, GitHub Actions (CI/CD), Docker, Microsoft Fabric, Azure, Visual Studio, VS Code, IntelliJ